europass

Gianluigi Rossi

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Website: https://github.com/galileo73/galileo73

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Gender: Male **Date of birth:** 04/05/1973 **Place of birth:** Roma, Italy **Nationality:** Italian

ABOUT ME

With over 20 years of experience as a Principal Engineer, I have spent a decade contributing to the Galileo Programme, while also gaining expertise in Copernicus and Security Accreditation for GOVSATCOM. My background touches GNSS systems, Galileo ground segment engineering and operations, PRS, service management, complemented by a strong foundation in security governance, compliance, risk analysis, and procurement.

Throughout my career, I've worked across Galileo ground segment design, system qualification, validation, operations and service roll-out, launch preparations, PRS, and service performance monitoring. I collaborate closely with stakeholders, ensuring that projects meet Service Implementation Plans, Qualificationa and Validation, and industry standards.

Some of the highlights of my work include overseeing Galileo service operations, monitoring SLA/KPI compliance, managing anomalies, and contributing to service engineering, ground segment procurement, PRS/GSMC evolution, and security accreditation. I enjoy tackling complex technical challenges, keeping projects on track, and delivering results—even under high-pressure conditions.

At my core, I am a customer-focused professional who thrives on strategic planning, problem-solving, and executing projects efficiently. I've been recognized for my ability to handle multiple priorities, collaborate effectively, and consistently deliver high-quality results, whether working independently or within a team.

Beyond my technical career, I have a creative side—I'm a passionate musician, playing keyboard and singing with my band, "Lost in Space."

WORK EXPERIENCE

[01/01/2024 – Current]

Copernicus Payload Data Acquisition Processing Maintenance and Evolution Engineer

Exprivia | EUMETSAT, Germany

City: Prague | **Country:** Czechia | **Business or sector:** Professional, scientific and technical activities

- Support to Sentinel-6B launch preparation activities including new Payload Data Acquisition and Processing (PDAP) baseline deployment and maintenance.
- Payload Data Acquisition and Processing (PDAP) System requirements design review for implementing new features and manage obsolescence.
- Anomaly/Non-Conformance analysis and follow up resolution.
- Multi Mission Element Processing Infrastructure Architect to integrate new orchestrator element into the Sentinel-6 operational baseline.
- Support to E-COPS and B-COPS Security Risk Analysis.
- Support to Sentinel-3 OLCI L2 Re-processing On boarding milestone

[14/06/2024 – 31/10/2024]

GOVSATCOM Security and Accreditation Engineer (Call For Expert)

EUSPA

City: Prague | **Country:** Czechia

- Support accreditation documentation preparation for Initial Service milestone.
- Ensure compliance through detailed review of security requirements, specs, and system design.
- Analyze and address security-related anomalies and non-conformances promptly.

[01/04/2022 – 31/12/2023] **Galileo Ground Segment Operations Engineer**

Rhea System SA | EUSPA, Prague

City: Prague | **Country:** Czechia

- Led comprehensive evolution reviews for the Galileo Ground Segment, fostering seamless collaboration with key project stakeholders.
- Developed and executed strategic Ground Segment deployment plans and migration strategy, effectively mitigating risks to align precisely with the Programme Roadmap.
- Played a pivotal role in planning and evaluation meetings for new Galileo Ground Segment assets, ensuring strict compliance with Service Validation plans and managing risk-adjusted schedules.
- Collaborated with cross-functional teams to provide crucial input for Security Accreditation milestones, collecting and delivering documentation from primary stakeholders for the Accreditation dossier (GMS, TSP, GRSP).
- Reviewed and meticulously assessed contractual change proposals under the Galileo Service Operator framework contract to ensure alignment with project objectives.
- Supported Service Exploitation Reviews, assessing Service Improvements, and conducting impact assessments on Service Implementation plans.
- Contributed to GSOp Monthly Performance Reviews, Quarterly Service Performance Reviews, and Service Improvement Plan Reviews, including the assessment of associated Liquidated Damages.
- Served as a Technical Expert for key ESA System Design Authority qualification milestones (TRR, PTR, TRB, DRB_EA) and actively participated in SCTC preparation and execution.
- Acted as the primary interface for PRS IOC evolution baseline with GSMC-FR and KMF topics.
- Prepared detailed technical reports and schedule presentations, leveraging tools like ARTS, SDT, CMAT, DMS, DOORS, MS Project, and Microsoft Office for efficient project management.

Key Achievements:

- Achieved a 10% reduction in proposal costs through comprehensive contractual reviews.
- Optimized migration timelines, minimized deployment risks, and ensured alignment with the Service Declaration Program Roadmap and PRS IOC Declaration.
- Shaped the Operational Concept for targeted Ground Segment scenarios.

[01/04/2020 – 31/03/2022] **Galileo Operations Engineer**

Rhea System SA | EUSPA, Prague

City: Prague | **Country:** Czechia

- Supported Ground Segment Operational Validation processes through roll-out, emphasizing expertise in KMF and PRS aspects. Coordinated key operational activities with project stakeholders to ensure seamless integration and service verification of the new Ground Segment baseline.
- Provided essential support for Ground Segment System Design milestones (CDR, PDR), contributing to the qualification and acceptance activities for Galileo Ground Segment Infrastructure releases. Focused on operational aspects to guarantee effective service verification.
- Contributed significantly to the GSMC Service Validation plan, particularly in security monitoring and PRS service interface with GMS. Coordinated GSMC/GMS

service validation test campaign preparation, directly supporting test cases as a technical expert in PRS and SECMON.
- Leveraged technical expertise to support KMF special operations in new satellite launch preparations.
- Played a key role in preparing Service Exploitation Reviews (SER), evaluating Service Improvements, and conducting impact assessments for Anomaly Reports (AR) and Non-Conformance Reports (NCR).
- Participated in Monthly Performance Meetings with GSOp, assessing compliance with Service Contract Key Performance Indicators (KPIs) from the perspective of Galileo Operations.
- Expert in using project tools such ARTS, SDT, CMAT, DMS, DOORS, MS Project, and Office to streamline project management and documentation.

Key Achievements:

- Developed and maintained weekly dashboards to provide real-time updates to supervisors on operational anomalies, impacts on processes, service implementation roadmap risks, and deviations from the working schedule.
- Coordinated GSMC/GMS Service Validation preparation, participating actively and providing valuable insights during the service validation test campaign with stakeholders.

[01/07/2018 – 31/03/2020]

Galileo Ground Segment Engineer

Rhea System SA | EUSPA, Prague

City: Prague | **Country:** Czechia

- Actively contributed to the design evolution reviews (PDR, CDR, QR) for the Galileo Ground Segment (GMS, GCS, GSC), ensuring alignment with project milestones and baseline schedules. Monitored deployment plans for the Ground Mission and Control Segment, PRS/SECMON, and GSC domains, emphasizing risk-adjusted approaches to support Service Implementation Plans.
- Played a key role in initial OSNMA and HAS/CAS Service Validation meetings, reviewing OS, HAS/CAS Service Implementation Plan documents and requirements.
- Reviewed ESA system qualification documents, including the Qualification Status Report, VCD, RVM, and SOC (as designed and as built). Provided support for the qualification and acceptance review of the new Galileo Ground Segment Infrastructure baseline through to service roll-out.
- Provided support for the preparation of Service Exploitation Reviews (SER), evaluating Service Improvements, and conducting AR/NCR impact assessments on Service Implementation Plans.
- Monitored ESA contractual compliance with SLA/KPI for the Anomaly Review Board and System Non-Conformance Review Board.
- Served as a technical expert for G2G Evolution Phase B, conducting a comprehensive review of GSEG requirements, MRD, SvRD, and MOCD.
- Prepared detailed technical reports and presentations.

Key Achievements:

- Established a process for delivering weekly dashboards, providing real-time updates on System Non-Conformances and impact risk analysis on qualification, baseline schedules, and working schedules.
- Led key meetings on major evolutions and contractual SLA/KPIs in the Galileo Departement, ensuring alignment with project objectives.
- Authored a comprehensive cost-benefit analysis for the G2G Ground Segment Early B1 phase closeout milestone, contributing to informed decision-making.
- Supported drafting the Statement of Work for the procurement preparation of the Specific Contract for GSCv2.0, showcasing strong ground segment engineering knowledge.

[01/04/2015 – 30/06/2018]

Galileo KMF Manager

Spaceopal GmBH Germany

City: Munich | **Country:** Germany

- Managed two teams and held weekly meetings to coordinate nominal and secure operational tasks. Identified and resolved operational issues while strategically planning operational validation phases leading up to roll-out in OPE chain.
- Supervised Galileo Special Operations during critical phases like satellite launch preparation, in-orbit tests, PRS operations, and Galileo System Builds functional qualification.
- Acted as a liaison with stakeholders such as Thales, Airbus Defence and Space, EUSPA, and ESA for KMF classified operations, ensuring effective coordination, communication, and collaboration.
- Supported monitoring, steering, and timely delivery of Security Operational Procedures (SOPs) to enhance the validation process by engineering teams.
- Investigated anomalies, ensuring resolution with all stakeholders, and monitored compliance with contractual SLA/KPI for operational efficiency.
- Developed and updated System Levels Operational Products (SLOPs) for KMF service provisioning, covering Service Continuity, Planned, and Unplanned Handover scenarios.
- Delivered technical notes, plans, and status reports to management. Served as a technical expert for Galileo Second Generation Ground Segment Phase B.

Key Achievements:

- Improved communication between teams by implementing weekly and monthly meetings to address actions, manage obstacles, and share operational knowledge for enhanced team performance.
- Delegated tasks for coordinating operational validation plans for new KMF releases, streamlining processes for greater efficiency.
- Enhanced the validation chain planning meeting by optimizing input provision from stakeholders, fostering collaboration and improving input integration.
- Developed the initial SLOP for Service Continuity, Planned, and Unplanned Handover scenarios, establishing a comprehensive operational framework for KMF services.

[01/03/2013 – 30/03/2015]

Solution Architect for Galileo PRS

Altran Italia | Telespazio Rome

City: Rome | **Country:** Italy

- Defined high-level functional and non-functional requirements based on project needs. Facilitated the flow-down of these requirements to the Telespazio project team, ensuring alignment.
- Developed the solution test plan and test cases, ensuring comprehensive coverage of functional and performance aspects.
- Led the selection of the best technology stack for development, coordinating tasks with the Telespazio project team to create an initial mock-up solution.
- Provided crucial support to project management by issuing weekly reporting dashboards. Organized and conducted monthly progress meetings to report on project status.
- Coordinated the integration and testing phase in collaboration with the Telespazio project team, ensuring seamless collaboration and adherence to project timelines.
- Implemented minor evolutions for an in-house GNSS Constellation Visibility tool, enhancing functionality and performance.

Key Achievements:

- Designed a full OSPF mock-up for testing Galileo OS/PRS signals using pseudo-satellites, demonstrating a deep understanding of signal testing methodologies.

- Produced high-quality reports, dashboards, and presentations for progress meetings, showcasing effective communication and reporting skills.
- Coordinated teams effectively during the final integration of elements, ensuring a smooth and efficient integration process.

[01/02/2012 – 31/12/2012]

Software Developer for SENECA project

Altran Italia | Telespazio Rome

City: Roma | **Country:** Italy

- Led the development, integration, and testing of software modules for satellite navigation in the SENECA project. Proficiently encoded and decoded GPS/GALILEO navigation messages, GBAS, RTCM, RINEX, and SP3 files using C++ and MATLAB.
- Integrated software modules to calculate ionospheric and tropospheric delays for GPS and Galileo signals, enhancing the accuracy and reliability of satellite navigation services.
- Successfully integrated and tested software modules in a GNSS Constellation simulator, ensuring functionality and performance within the simulation environment.

Key Achievements:

- Reached a key milestone by integrating the GALILEO NeQuick ionospheric model into the Constellation simulator, significantly improving simulation capabilities and contributing to the project's success.

[01/11/2010 – 28/02/2012]

Tactical Model Algorithm Engineer for Medium Extended Air Defence System (MEADS)

Altran Italia | MBDA Rome

City: Roma | **Country:** Italy

- Contributed to the development of tactical algorithms for Battle Management Command, Control, Communications, Computers, and Intelligence (BMC4I) in the MEADS program. Supported the Command-and-Control System Engineering MBDA Department in defining and implementing tactical algorithms for military threat engagement.
- Developed and integrated IFF target algorithms using C++ for Sol-Air Moyenne-Portée/Terrestre (SAMP/T), enhancing air and missile defense capabilities.

Key Achievements:

- Optimized the primary algorithm for threat detection, tracking, and engagement, significantly reducing decision time. This improvement enhanced the overall efficiency and responsiveness of the MEADS program in addressing military threats.

[01/03/2008 – 30/09/2010]

HMI-TCMS Software Engineer On board equipment Alstom ETR600/610 New Pendolino train platform

Altran Italia | Alstom Transport Italy

City: Roma | **Country:** Italy

- Led the development of the Human-Machine Interface (HMI) software for the ETR600/610 train platform, focusing on implementing the Train Control and Management System (TCMS) using Matlab/Simulink.
- Provided crucial commissioning support for the ETR600/610 train platform, overseeing software implementation and validation. Ensured seamless integration and functionality during commissioning activities in both Italy and Switzerland.
- Conducted comprehensive Software Validation and Testing phases onboard the ETR600/610 train platform in Italy and Switzerland. Rigorously tested and validated software functionalities to guarantee optimal performance and adherence to specifications.

Key Achievements:

- Played a key role in the successful commissioning of the ETR600/610 train platform, ensuring the operational readiness of the developed TCMS software in both Italy and Switzerland.
- Implemented a robust Software Validation and Testing process onboard, ensuring the reliability and functionality of the HMI software under real operating conditions. This approach enhanced overall platform performance and safety.

[01/07/2007 – 31/12/2007]

Software Engineer for Tornado/Eurofighter flight simulator

Altran Italia | Leonardo

City: Roma | **Country:** Italy

- Contributed to the development of critical software modules for a radar simulator on a Unix-like platform using C++. Ensured the robustness and functionality of the simulator's software components.
- Played a pivotal role in reviewing System Requirements Analysis, providing valuable insights to refine the software development process. Defined and documented software requirements for the radar simulator, ensuring alignment with system specifications and objectives.

Key Achievements:

- Demonstrated keen analytical skills in reviewing System Requirements Analysis, contributing significantly to refining and optimizing the software development process for the radar simulator.
- Successfully defined and documented accurate software requirements, ensuring the software components aligned with system specifications, contributing to the efficiency and effectiveness of the simulation software.

[01/01/2006 – 30/03/2007]

Software Engineer for GEOMOIST Project

MEEO S.r.l.

City: Roma | **Country:** Italy

- Led the modeling and implementation of advanced soil moisture retrieval algorithms, utilizing data from remote sensing platforms such as Envisat, MODIS, and Meteosat Second Generation. Focused on developing a robust Decision Support System (DSS) for comprehensive environmental monitoring.

Main Achievements:

- Contributed to the international scientific paper, "Estimating Soil Moisture Using Optical and Radar Satellite Remote Sensing Data," published in the NATO Science for Peace and Security Series C: Environmental Security in 2009. The paper harnessed project data on desertification and risk analysis, highlighting expertise in applying advanced remote sensing techniques for environmental studies.
- The paper, available at this link http://link.springer.com/chapter/10.1007%2F978-1-4020-8937-4_9) showcases significant contributions to the field of soil moisture estimation through the integration of optical and radar satellite remote sensing. This work underscores the crucial role of innovative remote sensing algorithms in enhancing soil moisture monitoring for environmental risk assessments.

[01/02/2005 – 31/12/2005]

Software Engineer for ESA DesertWatch project

ENEA C.R. Casaccia | ESA ESRIN

City: Roma | **Country:** Italy

- Contributed as a key member of the technical team developing a sophisticated Decision Support System (DSS) software. Applied advanced classification

techniques to analyze and interpret satellite images, enhancing the system's decision-making capabilities.

Main Achievements:

- Played an integral role in generating scientific knowledge through the publication of research papers based on project data. Presented these findings at prominent Italian Remote Sensing conferences, showcasing expertise in advanced classification techniques and their application in enhancing the DSS's functionality.

[01/06/2004 – 31/07/2004]

Software Engineer for DIPS project

Advanced Computer Systems A.C.S. Srl

City: Roma | **Country:** Italy

- Led the development of specialized Supervised Classification software for satellite image processing, executed on Silicon Graphics workstations. Focused on enhancing the accuracy and efficiency of satellite imagery classification.

Main Achievements:

- Spearheaded the creation of robust Supervised Classification software tailored for satellite image processing on Silicon Graphics workstations, significantly improving accuracy and efficiency in image classification.
- Played a crucial role throughout the software development lifecycle, ensuring seamless integration and conducting rigorous testing to meet stringent quality standards.

EDUCATION AND TRAINING

[2021]

PMI® Authorised PMP® Exam Prep

RHEA System SA , PMI, Learning Tree International https://www.pmi.org

City: Prague | **Country:** Czechia |

Leadership skills & techniques ■ Communicate effectively ■ Create an effective project Schedule ■

Project Risk Management ■ Emotional intelligence ■ Decision making ■ Conflict Management ■ Agile

project management

[2017]

ITIL awareness and GSOp Service Management Training Course

Spaceopal GmBH

City: Munich | **Country:** Germany |

Service delivery and customer satisfaction ■ Reduced costs through improved utilization of resources

■ Better management of business risk and service disruption ■ Stable service environment to support

constant business change

[2009]

Fundamentals of Communication, Time Management Training Course

Altran Italia

City: Roma | **Country:** Italy |

Listening ■ Communicate effectively ■ Public speaking ■ Collaborative mindset ■ Achieve goals ■

Reduce the stress ■ Prioritize the work

[2007]

Java/J2EE Certification

Axa Sistemi – Crismatica Consulting

City: Roma | **Country:** Italy |

[2004 – 2007]

Winner of PhD Contest in Computer Engineering and Science at the CINSA of Cagliari - XX ° Ciclo

University of Cagliari (Italy)

City: Roma | **Country:** Italy |

"The Delphi SM2" - Innovative Soil Moisture Retrieval Methodology in the Mediterranean Basin

Duration: Three years

Focus Area: Engineering and Environmental Science

Overview:

• Conducted an extensive three-year research project on "The Delphi SM2," exploring an innovative soil moisture retrieval methodology in the Mediterranean Basin.

Key Contributions:

• Developed and implemented an advanced soil moisture retrieval technique utilizing both active and passive remote sensing technologies.

Technical Implementations:

• Implemented software modules for soil moisture estimation using C++ programming language.

• Created Simulink models for ASAR sensor, MODIS, and Meteosat 2nd Generation, demonstrating proficiency in Simulink modeling.

[2004]

Master's Degree in Telecommunication Engineering

Università La Sapienza di Roma, Italy

City: Roma | **Country:** Italy | | **Thesis:** Integral Numerical Techniques for the Analysis of Dispersive Properties of Transmission Lines

Tools:

• Matlab, Simulink, and Fortran language for in-depth analysis.

Outcomes:

• Developed specialized software modules in Fortran for analyzing the propagation of the electromagnetic field in RF equipment.

• Conducted comprehensive data analysis using Matlab.

Skills Acquired:

• Gained expertise in Matlab and Simulink coding for effective data analysis and modeling.

• Proficient in Fortran coding, contributing to the development of specialized software modules for electromagnetic field analysis in RF equipment.

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

French

LISTENING A1 **READING** A1 **WRITING** A1

SPOKEN PRODUCTION A1 **SPOKEN INTERACTION** A1

Czech

LISTENING A1 **READING** A1 **WRITING** A1

SPOKEN PRODUCTION A1 **SPOKEN INTERACTION** A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

COMMUNICATION AND INTERPERSONAL SKILLS

Clear Communication, Adaptability , Technological Proficiency, Active Listening, Stakeholder Management , Negotiation

Demonstrated mastery in conveying complex information through clear and concise written

and verbal communication, adapting to various audiences and contexts

• Proficient in leveraging various communication technologies, ensuring effective

communication in remote or virtual work settings, and facilitating seamless collaboration

• Strong active listening skills, facilitating a deeper understanding of others' perspectives and

contributing to thoughtful and informed responses

• Display adaptability in communication style, tailoring approaches to diverse audiences and

situations

• Successfully managed communication with diverse stakeholders, including executives, and

team members, ensuring clarity and alignment with project objectives

• Skilled in negotiation communication, balancing assertiveness and cooperation to reach

mutually beneficial agreements in professional interactions

MANAGEMENT AND LEADERSHIP SKILLS

Multi-Project Management, Strategic Planning and Problem-Solving, Leadership, Time Management, Decision-Making

Successfully operated in multi-customer, multi-contract projects, showcasing the ability to

navigate complexities and ensure stakeholder satisfaction

• Demonstrated excellence in strategic problem-solving, utilizing analytical skills and experience

to effectively address challenges and contribute to project success.

• Possess strong strategic planning skills, contributing to the development and execution of

comprehensive plans aligned with organizational objectives

• Proven leadership capabilities of managing, inspiring and motivating teams to deliver highquality results aligned with organizational objectives

• Skilled in managing relationships with diverse stakeholders, ensuring effective communication

and collaboration to meet project requirements and exceed expectations

• Known for being a highly focused professional with a proven ability to meet and exceed

deadlines, consistently delivering results through effective time management and prioritization

• Exhibit adaptability and flexibility in navigating dynamic project environments, adjusting

strategies as needed to address evolving challenges and capitalize on emerging opportunities

• Display sound judgment and strategic decision-making skills, particularly in critical situations,

contributing to the overall success and sustainability of projects

DIGITAL SKILLS

My Digital Skills

Office Suite | GIT version control, Linux Command | Linux (Red Hat, Debian) | Python Language - Basic knowledge | Devops: Docker, Jenkins | Netbeans, Codeblocks, Kdevelop, Eclipse, and Visual Studio | C/C++, Java (Certified), Python and Javascript | Septentrio RX control, ERDAS Image, ENVI, and basic SCOS2000 | Atlassian stack (Jira, Confluence, ...) | Matlab/Simulink | DOORS | ECSS and MIL-STD-498 | Linux | ITIL | ARTS | NCTS | DMS | SDT | CMAT

HOBBIES AND INTERESTS

Musician, Composer, and Singer. I play keyboards, I do live performances on the stage I like publishing my music on major digital music stores (Spotify, Apple Music, Amazon Music, YouTube)

DRIVING LICENCE

Cars: B

PUBLICATIONS

[2009] **Estimating Soil Moisture Using Optical and Radar Satellite Remote Sensing Data**

Reference: NATO Security through Science Series C: Environmental Security

[2006] **A geostatistical method by means of remote sensing data for analysing landscape growth**

Reference: MondoGIS N°50

[2005] **Landscape evolution study applied to the Sinni's river (Basilicata) by processing remote sensing data and geostatistical analysis**

Reference: 7th Conference MondoGIS Geocorsi and training

[2005] **Confronto di procedure automatiche di classificazione con elevato numero di classi da immagini a bassa risoluzione**

Reference: Conference: IX Conferenza Nazionale ASITA

[2005] **Valutazione della capacita' di rigenerazione di aree incendiate attraverso indici di vegetazione ed analisi variografica**

Reference: Conference: IX Conferenza Nazionale ASITA